

Original Article – Editor's choice

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CaboPoint: A Phase 2 Study of Second-line Cabozantinib After Checkpoint Inhibitor–based Combination Therapy in Patients with Metastatic Renal Cell Carcinoma

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Abstract

Background and objective: Data are limited regarding pure second-line cabozantinib use after standard first-line checkpoint inhibitor (CPI)-based combination therapy for advanced renal cell carcinoma (aRCC). We report the final results from the prospective CaboPoint study.

Methods: The phase 2, multicentre, open-label CaboPoint study (NCT03945773) evaluated the efficacy and safety of second-line cabozantinib in adults with aRCC, after prior ipilimumab plus nivolumab (IpiNivo; cohort A) or CPI therapy plus vascular endothelial growth factor-targeted therapy (cohort B). The primary endpoint was objective response rate (ORR) as per Response Evaluation Criteria in Solid Tumors 1.1, evaluated by independent central review (ICR) in cohort A. The secondary endpoints included ORR by ICR in cohort B, ORR by investigator review, progression-free survival, overall survival, and safety.

Key findings and limitations: Overall, 127 patients were enrolled from 2020 to 2023 across 40 institutions. The median follow-up was 19.3 mo. ORRs (95% confidence interval [CI]) by ICR were 40.5% (29.6–52.1%) in cohort A (statistically significant and clinically meaningful) and 27.5% (14.6–43.9%) in cohort B; according to the investigator review, ORRs were 49.4% (38.4–60.5%) in cohort A and 33.3% (19.6–49.5%) in cohort B. The median (95% CI) progression-free survival was 10.9 (8.2–14.2) mo in cohort A and

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8.3 (5.6–11.1) mo in cohort B. The median (95% CI) overall survival was similar between cohorts (24.3 [18.5–31.8] and 24.1 [17.1–not calculable] mo, respectively). There were no new safety concerns.

Conclusions and clinical implications: CaboPoint is the first study of pure second-line cabozantinib after first-line CPI-based combinations, providing a benchmark for future second-line aRCC studies.

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ADVANCING PRACTICE

What does this study add?

CaboPoint is the first European, phase 2, open-label, prospective trial to evaluate second-line cabozantinib in patients with locally advanced or metastatic renal cell carcinoma after disease progression on first-line checkpoint inhibitor (CPI)-based combination therapy. Cabozantinib was efficacious with no new safety signals, which supports its continued use as second-line monotherapy in patients who progress on standard-of-care first-line CPI-based therapies.

Clinical Relevance

The results of the CaboPoint study offer a clear perspective on the activity of cabozantinib, a dual VEGF/MET inhibitor, after contemporary front-line therapies. Understanding the activity of the agent after dual immunotherapy and combination targeted therapy/immunotherapy approaches is of importance to the practicing clinician. Associate Editor: Sumanta Pal, MD.

Patient Summary

This study analysed how patients with advanced kidney cancer responded to cabozantinib after their disease worsened on their first treatment with checkpoint inhibitor (CPI)-based combination therapy. Cabozantinib worked well in patients who had previously received CPI-based therapies. These results confirm that cabozantinib is a standard of care as a second treatment for patients with advanced kidney cancer.

1. Introduction

In Europe, cabozantinib is indicated as first-line (1L) therapy for advanced renal cell carcinoma (RCC; as monotherapy for patients with intermediate or poor risk, or combined with nivolumab for all risk groups) and as second-line (2L) monotherapy following vascular endothelial growth factor (VEGF)-targeted treatment [1].

The 2L indication was based on the phase 3 METEOR study (NCT01865747), which reported significantly increased overall survival (OS), progression-free survival (PFS), and objective response rate (ORR) with cabozantinib versus everolimus, in patients with advanced clear-cell RCC (ccRCC) after prior VEGF-targeted tyrosine kinase inhibitor (TKI) therapy [2].

Since 2018, checkpoint inhibitor (CPI)-based therapies have become 1L standard-of-care therapy in advanced ccRCC [3,4]. Current European Association of Urology and European Society of Medical Oncology recommendations are CPI-VEGF-targeted therapy (cabozantinib + nivolumab, axitinib + pembrolizumab, and lenvatinib + pembrolizumab) and CPI-CPI therapy (ipilimumab + nivolumab [IpiNivo]) [5,6].

The phase 3 CONTACT-03 (NCT04338269; cabozantinib + atezolizumab) and phase 2 CANTATA (NCT03428217;

cabozantinib + telaglenastat) trials provided efficacy data on 2L and later-line (2L+) cabozantinib monotherapy (control arm in both trials) after prior CPI-based therapy [7,8]. However, neither trial focused on 2L cabozantinib only; instead, they enrolled patients who had received prior CPI. Third-line (3L) cabozantinib was used for many included patients (control arms: CONTACT-03, $n = 116/259$; CANTATA, $n = 93/223$). Additionally, not all patients in these trials received 1L combination therapy. Of note, response to 2L+ cabozantinib was similar in both monotherapy and combination arms.

No prospective trial has evaluated 2L cabozantinib after disease progression on 1L CPI combinations. The CaboPoint study aimed to address this knowledge gap.

2. Patients and methods

2.1. Study design

CaboPoint (NCT03945773) was a phase 2, multicentre, open-label study conducted (from January 8, 2020 to November 13, 2023) in Austria, France, Germany, Spain, Switzerland, the Netherlands, and the UK [9].

Enrolled patients had received either 1L CPI-CPI therapy (cohort A) or 1L CPI-VEGF-targeted therapy (cohort B) [9].

Patients received 2L cabozantinib 60 mg/d orally; investigators could reduce the dose to 40 or 20 mg/d to manage toxicity. Treatment continued until disease progression, unacceptable toxicity, consent withdrawal, or study end (maximum 18 mo after the last patient had initiated cabozantinib), whichever occurred first. Patients who discontinued treatment were followed every 12 wk until death or study end, whichever occurred first. Study exposure (time between date of informed consent and last visit) was used as a proxy for follow-up.

The study was conducted in accordance with the provisions of the Declaration of Helsinki and the International Conference on Harmonisation Consolidated Guidance on Good Clinical Practice. Study materials were reviewed and approved by relevant independent ethics committees/institutional review boards. All patients provided written informed consent prior to study entry.

2.2. Patient population

The eligibility criteria have been published previously [9].

2.2.1. Sample size

Cabozantinib was hypothesised to demonstrate a clinically significant increase in response rate compared with historical controls in 2L treatment [1]. Therapy was considered clinically meaningful if it provided a significant benefit in ORR as per an independent central review (ICR) over a standard rate of 10% based on the everolimus response rate in the METEOR study (3%; exact 95% confidence interval [CI]: 1.7–5.9). Owing to enrolment underperformance, sample size calculations were amended: 74 patients in cohort A would provide $\geq 80\%$ power (at a one-sided significance level [α] of 0.025) to reject the null hypothesis of 10% ORR in favour of an alternate hypothesis (ORR of 23%). For cohort B, no formal sample size was calculated; enrolment stopped when cohort A completed recruitment.

2.3. Outcomes

The primary endpoint was ORR (defined as a documented best overall response of a partial or complete response as per Response Evaluation Criteria in Solid Tumors [RECIST] 1.1) in cohort A, as per the ICR of patients with baseline target lesion and tumour assessment. Confirmed ORRs by ICR and local investigator review (LIR) were retrospectively analysed as an exploratory endpoint.

Secondary efficacy endpoints were time to response (TTR), duration of response (DOR), disease control rate, and PFS as per RECIST 1.1 by ICR and LIR; ORR as per RECIST 1.1 in cohort B by ICR; OS; ORR by LIR; and a quality-of-life (QoL) analysis (change in disease-related symptoms assessed using the Functional Assessment of Cancer Therapy–Kidney Symptom Index–Disease-related Symptoms [FKSI-DRS] questionnaire). TTR was defined as the time from study treatment initiation to the date of evidence of first response. Safety and tolerability were assessed throughout the study treatment period. Adverse events (AEs) were coded according to the most recent version of the Medical Dictionary for Regulatory Activities at study ini-

tiation and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events (CTCAE version 5.0).

2.4. Statistical analyses

2.4.1. Primary endpoint analysis

For cohort A, ORR by ICR was estimated and tested against a threshold of 10% (given the lack of prospective data) using a one-sample exact test for binomial distribution. Clinically meaningful benefit was defined as $>10\%$ improvement in ORR based on a p value for the one-sided test of <0.025 . ORR is presented with an associated two-sided 95% CI using the Clopper–Pearson exact method.

2.4.2. Analysis sets

All patients with one or more cabozantinib doses comprised the safety population; those who also had a baseline tumour assessment comprised the efficacy population.

The [Supplementary material](#) provides additional methodology, including treatment adherence, prespecified and post hoc subgroup analyses, and secondary endpoint analyses.

3. Results

3.1. Patient disposition

Of the 148 individuals screened, 127 were enrolled from 40 centres ([Supplementary Fig. 1A](#)). Treatment was discontinued in 113 patients, mostly due to radiological disease progression (61/127 [48.0%]) or AEs (23/127 [18.1%]). Of the 54 patients who completed the study, 14 received treatment till the end of the study. Among 73 patients who discontinued prematurely, there were 68 on-study deaths. [Supplementary Fig. 1B](#) summarises the analysis sets.

3.2. Demographics and clinical characteristics

Among 127 enrolled patients, the mean age was 64.2 yr and 75.6% were male ([Table 1](#)). Most patients (65/75 [86.7%]) had International Metastatic RCC Database Consortium (IMDC) intermediate or poor risk at diagnosis; 52 patients had missing data. Baseline characteristics were generally balanced between cohorts A ($n = 85$) and B ($n = 42$). A large proportion of patients had locally advanced (17.6%) or metastatic (61.6%) disease at diagnosis.

3.3. Efficacy

The median (interquartile range [IQR]) follow-up was 19.3 (10.9–26.3) mo: cohort A, 19.3 (11.2–26.5) mo; cohort B, 20.4 (10.3–24.3) mo. All patients comprised the efficacy population for investigator-reported outcomes. Eight patients (A, $n = 6$; B, $n = 2$) lacked a baseline central assessment of a target lesion. Therefore, 119 patients (A, $n = 79$; B, $n = 40$) comprised the efficacy population for ICR-reported outcomes.

The ORR (95% CI) was 40.5% (29.6–52.1%; 32/79) in cohort A (primary endpoint), 27.5% (14.6–43.9%; 11/40) in cohort B, and 36.1% (27.5–45.4%; 43/119) overall ([Fig. 1A](#)). The primary endpoint was statistically significant against

Table 1 – Baseline patient demographics and clinical characteristics (safety population; N = 127)

	Cohort A (n = 85)	Cohort B (n = 42)	Overall (N = 127)
Age (yr), mean (SD)	63.2 (9.0)	66.2 (9.7)	64.2 (9.3)
Sex (male), n (%)	63 (74.1)	33 (78.6)	96 (75.6)
ECOG PS, n (%)			
0	52 (61.2)	25 (59.5)	77 (60.6)
1	33 (38.8)	17 (40.5)	50 (39.4)
IMDC risk group, prognosis at diagnosis, n (%) ^a			
Favourable	3 (6.3)	7 (25.9)	10 (13.3)
Intermediate	36 (75.0)	15 (55.6)	51 (68.0)
Poor	9 (18.8)	5 (18.5)	14 (18.7)
Missing	37	15	52
Sum of measurements of all target lesions per patient (mm) ^b			
Mean (SD)	98.6 (60.6)	81.7 (60.5)	92.9 (60.8)
Median (IQR)	84.9 (53.4–122.0)	61.3 (37.1–113.7)	71.6 (45.1–118.1)
RCC stage at diagnosis, n (%) ^a			
Localised	12 (14.5)	14 (33.3)	26 (20.8)
Locally advanced	14 (16.9)	8 (19.0)	22 (17.6)
Metastatic	57 (68.7)	20 (47.6)	77 (61.6)
Missing	2	0	2
Prior nephrectomy for RCC, n (%)	49 (57.6)	32 (76.2)	81 (63.8)
Metastatic localisation at the start of cabozantinib treatment, n (%) ^{a,c}			
Lungs	59 (70.2)	30 (71.4)	89 (70.6)
Bones	22 (26.2)	14 (33.3)	36 (28.6)
Liver	15 (17.9)	7 (16.7)	22 (17.5)
Lymph nodes	36 (42.9)	19 (45.2)	55 (43.7)
Visceral, other	3 (3.6)	1 (2.4)	4 (3.2)
Brain	5 (6.0)	1 (2.4)	6 (4.8)
Other	29 (34.5)	12 (28.6)	41 (32.5)
Missing	1	0	1
Number of metastatic sites, n (%) ^a			
1	27 (32.1)	11 (26.2)	38 (30.2)
2	22 (26.2)	16 (38.1)	38 (30.2)
≥3	35 (41.7)	15 (35.7)	50 (39.7)
Missing	1	0	1
Duration of first-line treatment, n (%)			
Median (IQR)	5.0 (2.1–10.8)	8.6 (5.7–14.0)	6.4 (2.5–12.4)
<3 months	33 (38.8)	3 (7.1)	36 (28.3)
3–<6 months	15 (17.6)	10 (23.8)	25 (19.7)
≥6 months	37 (43.5)	29 (69.0)	66 (52.0)
Prior systemic therapies for RCC, n (%)			
Nivolumab + ipilimumab	84 (98.8)	0	84 (66.1)
Nivolumab ^d	1 (1.2)	0	1 (0.8)
Axitinib + avelumab	0	13 (31.0)	13 (10.2)
Axitinib + pembrolizumab	0	29 (69.0)	29 (22.8)

ECOG PS = Eastern Cooperative Oncology Group performance status; ICR = independent central review; IMDC = International Metastatic RCC Database Consortium; IQR = interquartile range; RCC = renal cell carcinoma; SD = standard deviation.

^a Percentages were based on the number of patients with nonmissing values.

^b Assessed by ICR; 119 patients had baseline lesions for assessment by a central review (79 in cohort A and 40 in cohort B).

^c Patients could have had more than one localisation.

^d Protocol deviation was reported for one patient in cohort A who had received nivolumab monotherapy.

the 10% threshold ($p < 0.0001$). Disease control rates were >80% in both cohorts and overall (Fig. 1A). In cohorts A and B, respectively, there were 32/76 (42.1%) and 10/37 (27.0%) partial responses, and zero of 76 (0%) and one of 37 (2.7%) complete responses. Patients with missing data (three per cohort) were considered nonresponders (patients

with progressive disease) for the ORR analysis to be conservative, but their best overall response data were not presented (Fig. 1A and B).

The ORR (95%) by investigator (secondary endpoint) was 49.4% (38.4–60.5%) in cohort A and 33.3% (19.6–49.5%) in cohort B (Fig. 1B).

Concordance of responses was 77.6% in cohort A and 73.0% in cohort B (see the [Supplementary Table 1](#)). The median best percentage reduction in baseline tumour size was <31.2% and, although no statistical comparisons were conducted, was numerically higher in cohort A than in cohort B, irrespective of the assessment type (see the [Supplementary results and Supplementary Fig. 1](#)).

PFS and DOR were numerically longer (no statistical comparison) in cohort A than in cohort B (Fig. 2 and [Supplementary Fig. 3](#)), while TTR and OS were similar between cohorts ([Supplementary results and Fig. 2](#)).

Despite small patient numbers, ORR by ICR was nominally higher (no statistical comparison) in the following patient subgroups for both cohorts: age <65 versus ≥65 yr, Eastern Cooperative Oncology Group performance status 0 versus 1, one to two versus three or more metastatic sites, IMDC favourable versus intermediate/poor risk; and prior nephrectomy versus without (for cohort A only; [Supplementary Fig. 4](#)).

Patients in both cohorts received subsequent therapies; 48.8% received 3L therapy. Of these patients, those in cohort A received up to six therapy lines and those in cohort B received up to four therapy lines (see the [Supplementary results and Supplementary Table 2](#)).

QoL as per FKSI-DRS score declined significantly by study end in cohort A ($p = 0.0019$), but not in cohort B ($p = 0.0596$; see the [Supplementary results and Supplementary Fig. 5](#)).

3.4. Post hoc analysis

Confirmed ORRs (response on two consecutive imaging assessments) by ICR were 34.2% (95% CI: 23.9–45.7%; 27/79) in cohort A, 20.0% (9.1–35.6%; eight of 40) in cohort B, and 29.4% (21.4–38.5%; 35/119) overall. Confirmed ORRs (95% CI) by LIR were 29.4% (20.0–40.3%; 25/85) in cohort A, 19.0% (8.6–34.1%; eight of 42) in cohort B, and 26.0% (18.6–34.5%; 33/127) overall.

3.5. Safety

Cabozantinib exposure, dose reductions, and dose interruptions were generally similar between cohorts (Table 2). The median (IQR) total cabozantinib exposure was 8.8 (4.8–17.5) mo: cohort A, 9.2 (4.8–19.8) mo; cohort B, 8.4 (5.0–14.8) mo. Among 93 patients with available data, 73 (78.5%) adhered to treatment; results were similar between cohorts ([Supplementary results](#)).

Most patients (99.2%) reported any-grade treatment-emergent adverse events (TEAEs), most of which were mild to moderate (Table 2). Grade 3–4 TEAEs were reported by 96 patients (75.6%) (cohort A, 71 [83.5%]; cohort B, 25 [59.5%]), most commonly hypertension (32.3%), diarrhoea (11.0%), and pulmonary embolism (7.9%; Table 2 and

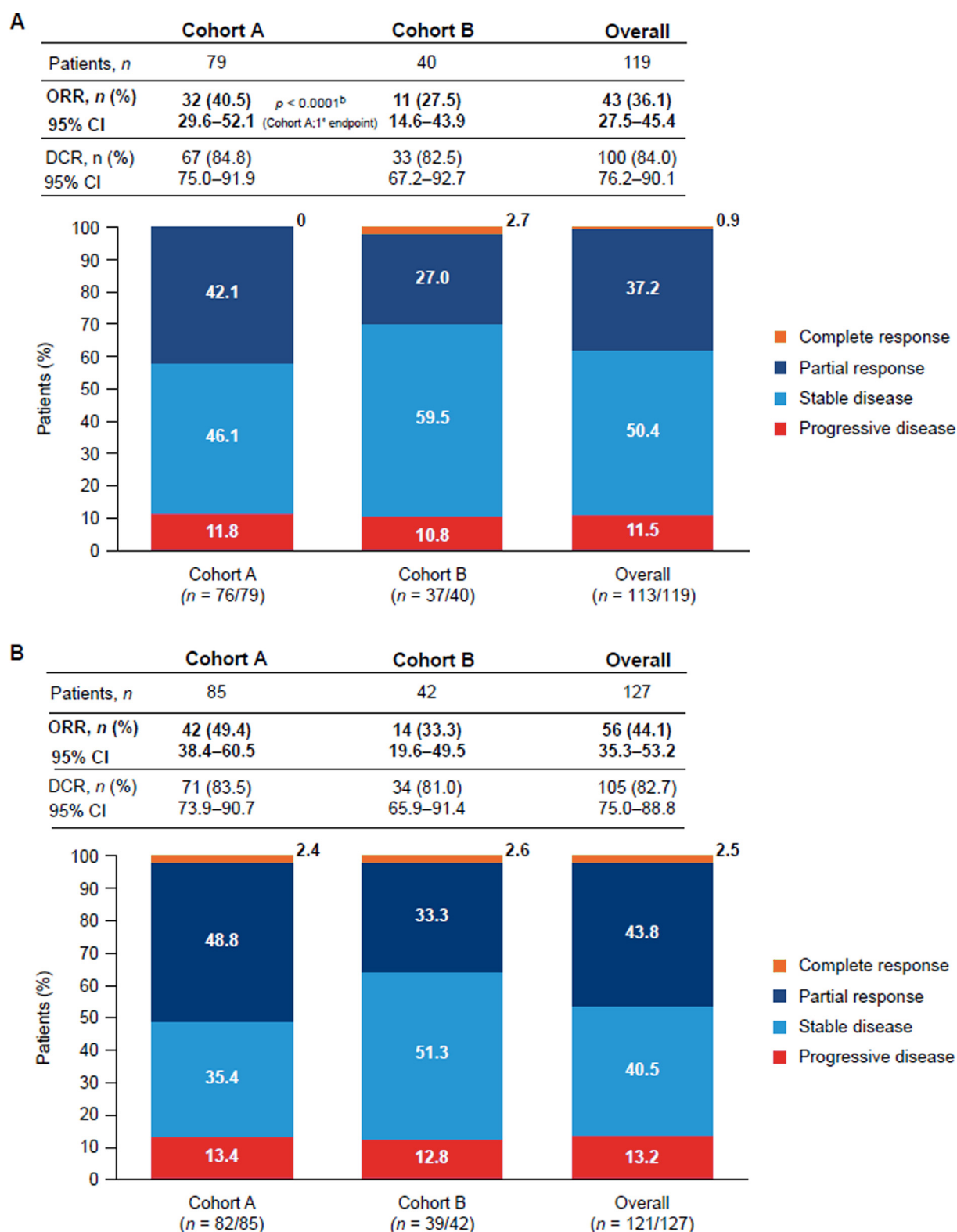


Fig. 1 – ORR and DCR by (A) ICR (*n* = 119^a) and (B) investigator (*n* = 127) in cohort A (prior CPI-CPI therapy) and cohort B (prior CPI-VEGF-targeted therapy). ORR is defined as the proportion of patients with a complete or partial response. DCR is defined as the proportion of patients with a complete response, a partial response, or stable disease. ORR and DCR percentages are based on the total number of patients with available data; those with missing data were considered nonresponders. BOR percentages are based on the number of patients with nonmissing data; patients with missing data are not presented for data assessed by ICR and IIR. BOR = best overall response; CI = confidence interval; CPI = checkpoint inhibitor; DCR = disease control rate; ICR = independent central review; IIR = local investigator review; ORR = objective response rate; VEGF = vascular endothelial growth factor. ^a Eight patients (six patients in cohort A and two patients in cohort B) were excluded from the primary analysis efficacy population because they lacked baseline target lesions for central tumour assessment. ^b Tested against the threshold of 10% (from the METEOR trial) using a one-sample exact test for binomial distribution.

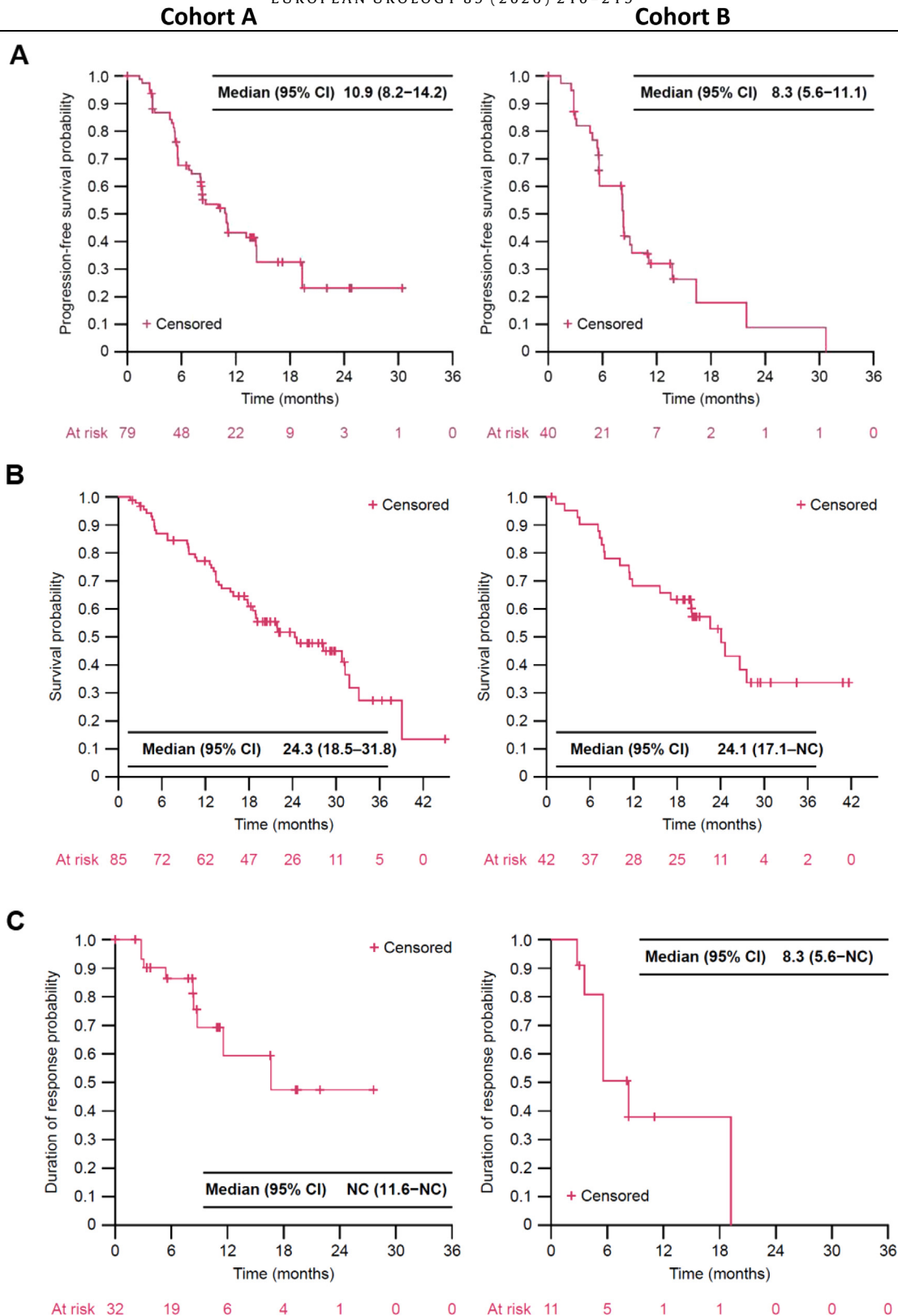


Fig. 2 – (A) PFS by ICR, (B) OS, and (C) DOR by ICR in cohort A (prior CPI-CPI therapy) and cohort B (prior CPI-VEGF-targeted therapy). PFS and DOR by ICR efficacy population: $n = 119$ (cohort A, $n = 79$; cohort B, $n = 40$). DOR is analysed for responders only (ie, 32/79 patients in cohort A and 11/40 patients in cohort B). OS efficacy population: $n = 127$ (cohort A, $n = 85$; cohort B, $n = 42$). PFS is defined as the time from the start of study treatment to either disease progression or death due to any cause, whichever occurred first. OS is defined as the time from the start of study treatment to the date of death. DOR is defined as the time from the first documented response to either disease progression or death due to any cause, whichever occurred first. Both PFS and DOR were censored at the last tumour assessment for those patients who were not known to have disease progression or be dead at data cut-off. Patients who (1) received nonprotocol anticancer therapy (ie, any cancer therapy that could have jeopardised the assessment of efficacy) after the start of cabozantinib (before an event) were censored at the date of the last tumour assessment prior to the date of initiating subsequent therapy, (2) did not have an event at data cut-off (and were not censored for any other reason) were censored the last tumour assessment prior to data cut-off, and (3) had an event after missing two or more scheduled tumour assessments were censored at the date of the last tumour assessment. In the overall population, median (95% CI) PFS was 9.0 (8.2–11.1) mo and OS was 24.3 (19.2–30.8) mo. PFS and DOR by investigator are presented in Supplementary Fig. 3. CI = confidence interval; CPI = checkpoint inhibitor; DOR = duration of response; ICR = independent central review; NC = not calculable; OS = overall survival; PFS = progression-free survival; VEGF = vascular endothelial growth factor.

Table 2 – Summary of safety (safety population; N = 127)

	Cohort A (n = 85)	Cohort B (n = 42)	Overall (N = 127)
Study drug exposure ^a			
Median (IQR), mo	9.2 (4.8–19.8)	8.4 (5.0–14.8)	8.8 (4.8–17.5)
Median (IQR) average daily dose (mg/d)	38.3 (27.4–50.0)	39.3 (30.4–50.9)	38.7 (27.5–50.9)
Dose reductions			
n (%)	26 (30.6)	17 (40.5)	43 (33.9)
Median (IQR) time to first reduction (d)	65.5 (29.0–141.0)	84.0 (57.0–113.0)	72.0 (29.0–141.0)
Dose interruptions			
n (%)	72 (84.7)	33 (78.6)	105 (82.7)
Median (IQR) time to first interruption (d)	30.0 (23.0–57.0)	33.0 (16.0–71.0)	30.0 (20.0–59.0)
Adverse events, n (%)			
Any TEAE	85 (100)	41 (97.6)	126 (99.2)
Any grade 3–4 TEAE ^b	71 (83.5)	25 (59.5)	96 (75.6)
Any serious AE	54 (63.5)	16 (38.1)	70 (55.1)
Any serious TEAE	51 (60.0)	15 (35.7)	66 (52.0)
Any TEAE leading to cabozantinib discontinuation	16 (18.8)	8 (19.0)	24 (18.9)
Any TEAE leading to cabozantinib interruption	75 (88.2)	32 (76.2)	107 (84.3)
Any TEAE leading to dose reduction of cabozantinib	26 (30.6)	11 (26.2)	37 (29.1)
Any TEAE with fatal outcome ^c	0	1 (2.4)	1 (0.8)

AE = adverse event; IQR = interquartile range; TEAE = treatment-emergent adverse event.

^a Defined as the time from the first to the last dose.

^b Based on Common Toxicity Criteria for Adverse Events severity.

^c No deaths were due to treatment-related TEAEs (see the [Supplementary results](#) for additional safety data).

Fig. 3. The median (95% CI) time to onset was 1.3 (0.9–1.9) mo overall (cohort A, 1.0 [0.6–1.8] mo; cohort B, 2.4 [0.9–not calculable] mo).

Most TEAEs were treatment related; the most common were diarrhoea (66.1%), decreased appetite (43.3%), hypertension (42.5%), and stomatitis (40.9%; [Supplementary Fig. 6](#)). Treatment-related TEAEs led to treatment interruption in 78.8% and 76.2%, and dose reduction in 30.6% and 26.2% of patients in cohorts A and B, respectively. Treatment-related serious TEAEs occurred in 25.2% of patients (cohort A, 29.4%; cohort B, 16.7%), most commonly diarrhoea and pulmonary embolism (four patients each). No deaths were due to treatment-related TEAEs (see the [Supplementary results](#) for additional safety data).

4. Discussion

CaboPoint was the first prospective study of cabozantinib after 1L CPI-based combination therapy; other studies included subgroups of CPI-pretreated patients [7,8]. At a median follow-up 19.3 mo, 2L cabozantinib was associated with a clinically meaningful ORR of 40.5% ($p < 0.0001$) after IpiNivo and a high ORR of 27.5% after CPI-VEGF-targeted therapy (axitinib plus avelumab or axitinib plus pembrolizumab). Numerically longer DOR and PFS may indicate greater response to 2L cabozantinib following prior IpiNivo (cohort A) compared with prior CPI-VEGF-targeted therapy (cohort B).

Consistent with side-effect management recommendations, particularly in older patients, most patients maintained cabozantinib below 60 mg/d (starting dose). Indeed, TEAEs led to cabozantinib interruption in >75% and reduction in >25%, but discontinuation in <20% of patients. Cabozantinib demonstrated benefit at lower doses, supporting findings from METEOR that response may be dose independent [10]. Despite a similar average daily cabozantinib dose to METEOR, discontinuation rate due to AEs was higher (19% vs 12% in METEOR) [2]. All TEAEs in CaboPoint were consistent with the known safety profile of cabozantinib, and rates were similar irrespective of 1L therapy. Observed rates of any-grade palmar-plantar erythrodysesthesia were higher in cohort A (45.9%) than in cohort B (23.8%), which may warrant further investigation in future studies. The grade 3–4 TEAE rate (75.6%) was consistent with those of METEOR (68%), and CANTATA (79%) and CONTACT-03 (62%) control arms [7,8]; there were no treatment-related deaths. Differences in grade 3–4 and serious TEAE rates between cohorts may indicate that tolerability of 2L cabozantinib is dependent on 1L therapy. Indeed, QoL was lower in cohort A patients who were TKI naïve and susceptible to associated side effects when compared with cohort B patients, who had TKI experience and may have managed their side effects more effectively (reflected by more dose reductions, fewer interruptions, and longer time to dose reduction). Differences in baseline IMDC risks between cohorts may also have led to between-cohort disparities in QoL, with IpiNivo (cohort A) being licensed for intermediate/poor-risk patients only.

Efficacy in CaboPoint was similar to that in CANTATA and CONTACT-03 cabozantinib monotherapy control arms and the recent real-world, retrospective, UK-based CARINA study [7,8,11]. In CONTACT-03, ORR (secondary endpoint) was 41% versus 41% in the cabozantinib + atezolizumab and cabozantinib monotherapy control arms. Response to 2L cabozantinib appears to be dependent on 1L therapy, with a higher ORR in cohort A than in cohort B, consistent with the findings of CARINA [11]. Indeed, ORRs with 2L cabozantinib in CARINA were 36.0% (41/114) after 1L CPI-CPI therapy (mostly IpiNivo) and 16.0% (4/25) after 1L CPI-TKI therapy [11]. This indicates that cabozantinib is an effective 2L treatment option following CPI-CPI and CPI-TKI, as does the similar OS in both CaboPoint cohorts. Differences in ORRs between cohorts may be due to prior exposure to another VEGF TKI and therefore cross-VEGF inhibitor resistance, or prior immune activation from dual checkpoint blockade.

Demographics were similar in cohort A and control arms in CONTACT-03 (cabozantinib following 1L or 2L CPI) [7] and CANTATA (cabozantinib following 1L or 2L antiangiogenic therapy or IpiNivo) [8]. Notable differences in therapies given before cabozantinib included the following: CPI therapy was 2L in CONTACT-03 and CANTATA versus 1L in CaboPoint, 63% of CONTACT-03 patients had prior VEGF-TKI therapy (monotherapy or combined with CPI) versus 0% in cohort A and 100% in cohort B, 100% of CANTATA patients had prior antiangiogenic therapy versus 0% in cohort A, and prior IpiNivo rate was low in CONTACT-03

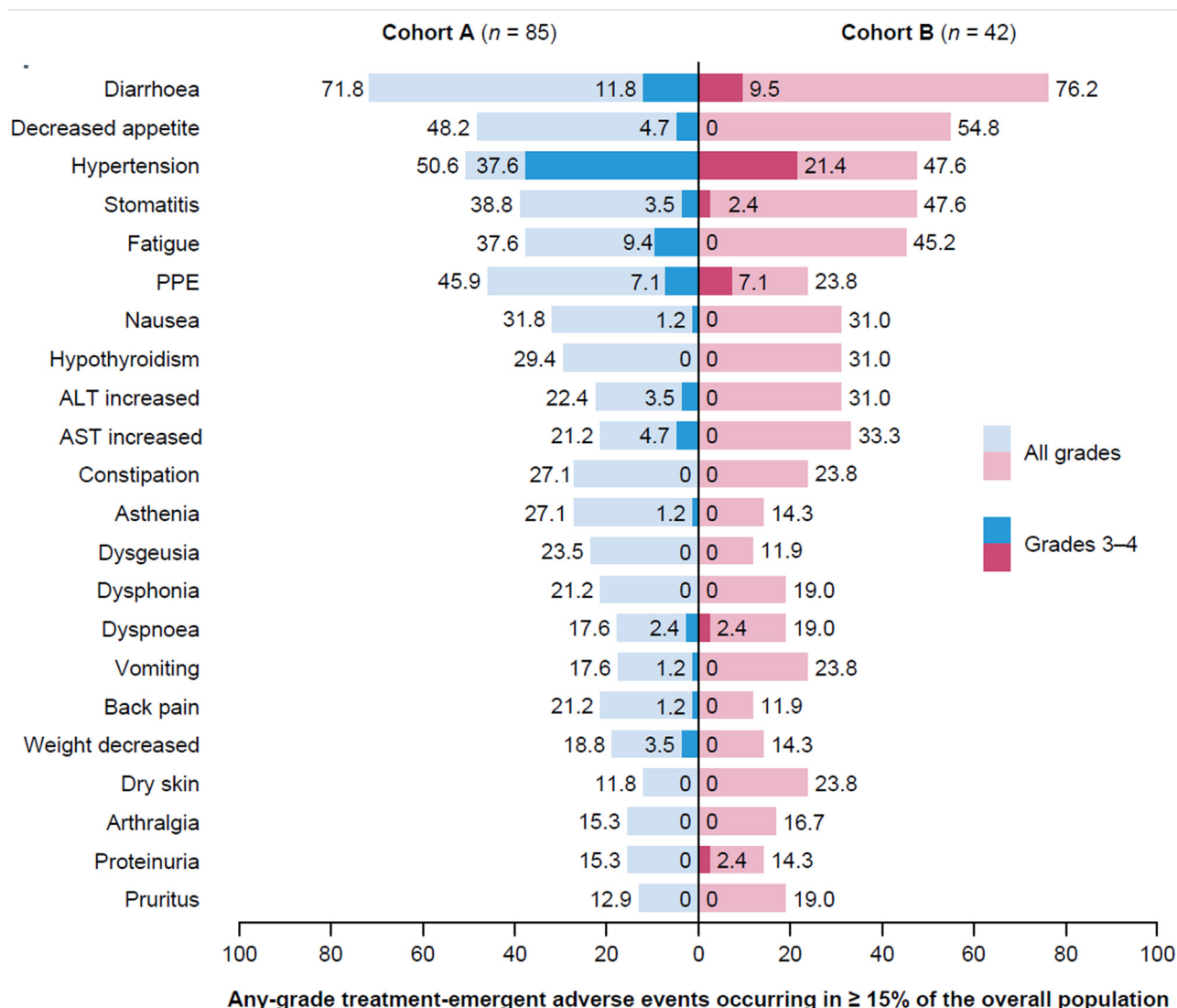


Fig. 3 – Most common TEAEs by grade (safety population; $N = 127$). ALT = alanine aminotransferase; AST = aspartate aminotransferase; PPE = palmar-plantar erythrodysesthesia; TEAE = treatment-emergent adverse event.

(27%) and CANTATA (29%) versus cohort A (98.8% [66.1% overall]) [7,8].

The prospective, multisite, international design of Cabo-Point enables generalisation of findings across Europe. Results reflect current RCC treatment guidelines and prescribing practices of TKI monotherapy after disease progression on 1L CPI-based therapy. Limitations included no head-to-head comparator, nonrandomised open-label design, sample size amendment due to enrolment underperformance, and missing IMDC and subgroup category information. Findings may not be generalisable to populations outside Europe and other real-world settings where patient characteristics and health care infrastructures differ.

Whether to use monotherapy or combination therapy after 1L CPI-based combination therapy is under investigation in the on-going phase 3 LITESPARK-011 study (NCT04586231), which will compare the outcomes of 2L

belzutifan (hypoxia-inducible factor-2 alpha inhibitor) plus lenvatinib (TKI) combination therapy with those of 2L cabozantinib in patients with advanced RCC [12].

Our findings pose several interesting questions: What are the main demographic and clinical characteristic differences between patients with and without dose reduction? How is treatment response affected by cabozantinib dose intensity and dose modifications [13,14], or clinical/disease characteristics (hypertension, nephrectomy, metastatic organ site, and IMDC risk group)? What is the relationship between body composition and the response to and toxicity of cabozantinib? How does QoL change throughout 2L cabozantinib treatment? Future research into biomarker-driven treatment approaches may help optimise 2L therapy selection by identifying which patients will best respond to cabozantinib, as monotherapy or with nivolumab.

5. Conclusions

This European, phase 2, open-label study demonstrated the efficacy of pure 2L cabozantinib in patients with advanced RCC after CPI-based combinations. No new safety signals were reported. Our findings support the continued use of 2L cabozantinib monotherapy in patients whose disease progresses on 1L CPI-based therapies and serves as a benchmark for future 2L trials.

Author contributions: Laurence Albiges had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

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Supplementary material

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References

- [1] Cabometyx. Summary of product characteristics. www.ema.europa.eu/en/documents/product-information/cabometyx-epar-product-information_en.pdf.
- [2] Choueiri TK, Escudier B, Powles T, et al. Cabozantinib versus everolimus in advanced renal cell carcinoma (METEOR): final results from a randomised, open-label, phase 3 trial. *Lancet Oncol* 2016;17:917–27.
- [3] Powles T, ESMO Guidelines Committee. Recent eUpdate to the ESMO clinical practice guidelines on renal cell carcinoma on cabozantinib and nivolumab for first-line clear cell renal cancer: renal cell carcinoma: ESMO clinical practice guidelines for diagnosis, treatment and follow-up. *Ann Oncol* 2021;32:422–3.
- [4] Ljungberg B, Albiges L, Abu-Ghanem Y, et al. European Association of Urology guidelines on renal cell carcinoma: the 2022 update. *Eur Urol* 2022;82:399–410.
- [5] Powles T, Albiges L, Bex A, et al. Renal cell carcinoma: ESMO clinical practice guideline for diagnosis, treatment and follow-up. *Ann Oncol* 2024;35:692–706.
- [6] European Association of Urology 2024. Guidelines on renal cell carcinoma. <https://d56bochluxqnz.cloudfront.net/documents/full-guideline/EAU-Guidelines-on-Renal-Cell-Carcinoma-2024.pdf>.
- [7] Pal SK, Albiges L, Tomczak P, et al. Atezolizumab plus cabozantinib versus cabozantinib monotherapy for patients with renal cell carcinoma after progression with previous immune checkpoint inhibitor treatment (CONTACT-03): a multicentre, randomised, open-label, phase 3 trial. *Lancet* 2023;402:185–95.
- [8] Tannir NM, Agarwal N, Porta C, et al. Efficacy and safety of telaglenastat plus cabozantinib vs placebo plus cabozantinib in patients with advanced renal cell carcinoma: the CANTATA randomized clinical trial. *JAMA Oncol* 2022;8:1411–8.
- [9] Albiges L, Schmidinger M, Taguieva-Pioger N, Perol D, Grunwald V, Guemas E. CaboPoint: a phase II study of cabozantinib as second-line treatment in patients with metastatic renal cell carcinoma. *Future Oncol* 2022;18:915–26.
- [10] Donskov F, Motzer RJ, Voog E, et al. Outcomes based on age in the phase III METEOR trial of cabozantinib versus everolimus in patients with advanced renal cell carcinoma. *Eur J Cancer* 2020;126:1–10.
- [11] Nathan P, Venugopal B, Ali J, et al. Real-world treatment sequencing and outcomes with cabozantinib after first-line immune checkpoint inhibitor-based combination therapy for patients with advanced renal cell carcinoma: CARINA study results. *Clin Genitourin Cancer* 2024;22:102141.
- [12] Motzer RJ, Schmidinger M, et al. LITESPARK-011: belzutifan plus lenvatinib vs cabozantinib in advanced renal cell carcinoma after anti-PD-1/PD-L1 therapy. *Future Oncol* 2023;19:113–21.
- [13] Thibault C, Mourey L, Pasquier D, et al. OCTOPUS: A retrospective, multicenter study of real-world cabozantinib treatment of patients with advanced renal cell carcinoma in France—subgroup analyses of treatment sequences and optimization. *J Clin Oncol* 2025;43:472.
- [14] Albiges L, Fléchon A, Chevreau C, et al. Real-world evidence of cabozantinib in patients with metastatic renal cell carcinoma: results from the CABOREAL early access program. *Eur J Cancer* 2021;142:102–11.